

YULIANG FU

North Carolina State University ◊ Raleigh, NC, USA
yfu34@ncsu.edu ◊ [Google Scholar](#)

EDUCATION

North Carolina State University, Raleigh, NC

Ph.D. in Computer Science (*In Progress*)

Expected May 2029

M.S. en route in Computer Science

May 2026

Tsinghua University, Beijing, China

Dual **B.S.** in Mathematics and Physics; Electrical Engineering and Automation

July 2024

RESEARCH INTERESTS

Ubiquitous Computing, Human-Computer Interaction, Wearable and Wireless Sensing, Mobile Health, Multimodal Signal Processing, Deep Learning for Sensing Systems.

RESEARCH EXPERIENCE

SSCS Lab, North Carolina State University, Raleigh, NC

Aug 2024 – Present

Ph.D. Student, advised by Dr. [Chenhan Xu](#)

- Develop wearable and wireless sensing systems for interactive computing and daily health monitoring using mmWave radar, bioacoustics, and IMUs. Explore few-shot learning for data with high interpersonal variability.
- First-authored *EpiPad* at IMWUT/UbiComp 2026 and *TouchWave* at IEEE BSN 2025. Contributed to multiple other publications on activity monitoring and bioacoustic sensing.

SciFi Lab, Cornell University, Ithaca, NY

Jun 2023 – Sep 2023

Student Research Intern, advised by Dr. [Cheng Zhang](#)

- Built an active-acoustic facial expression tracking system using commodity headsets. Optimized deep learning pipeline for acoustic spectrogram feature extraction.

Pervasive HCI Group, Tsinghua University, Beijing, China

Apr 2022 – Jun 2023

Research Assistant, advised by Dr. [Yuntao Wang](#)

- Developed an FMCW acoustic-ranging method for home AIoT devices. Built an ad hoc network for low-latency sensing system communications.

PROFESSIONAL EXPERIENCE

Research and Development Intern, JD.com, Beijing, China

Nov 2023 – Jan 2024

- Developed a task redistribution and workload analysis system using logistics terminals' IMU data.

PUBLICATIONS

[4] **Yuliang Fu**, Rakshita Ranganath, Huining Li, and Chenhan Xu, "*EpiPad: Enabling Multi-Finger and Force-Sensitive Skin Interface for Smartwatches*", Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (ACM IMWUT/UbiComp 2026).

[3] **Yuliang Fu**, Zhi Zhang, Rakshita Ranganath, Zhizhen Li, Yuchen Liu, Ning Sui, Huining Li, Chenhan Xu, "*TouchWave: Exploring mmWave-based Non-contact Fingertip-force Sensing in Activities of Daily Living*", IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025)

[2] Zhi Zhang, Chongxin Zhong, **Yuliang Fu**, Ying Chen, Jinyuan Jia, Huining Li, Chenhan Xu, "*mV-IMU: mmWave-enabled Virtual Inertia Measurement Unit for High-fidelity Activities of Daily Living Monitoring*", IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025)

[1] Rakshita Ranganath, Zhi Zhang, **Yuliang Fu**, Jinyuan Jia, Ying Chen, Huining Li, Chenhan Xu, “*Hand-Grip Strength Estimation through Bioacoustic Sensing*”, IEEE-EMBS International Conference on Body Sensor Networks (BSN 2025)

AWARDS & HONORS

- Graduate Merit Award, North Carolina State University 2026
- GSA Travel Assistance Award, North Carolina State University 2025
- Student Travel Award, IEEE-EMBS Conference on Body Sensor Networks 2025

SERVICES

Reviewer

- IEEE Journal of Biomedical and Health Informatics (JBHI)
- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)
- International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) 2025
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) 2025, 2026

Technical Program Committee Member

- IEEE-EMBS International Conference on Body Sensor Networks (BSN) 2025, 2026

TECHNICAL SKILLS

Programming	Python, C/C++, SQL, MATLAB
ML/Data	PyTorch, scikit-learn, NumPy, SciPy, pandas
Sensing and Hardware	mmWave radar, Acoustic/Ultrasonic sensing, IMUs, Nordic nRF52, STM32
Tools & Methods	Git, Linux, LaTeX, Signal Processing, Sensor Fusion, Experimental Design